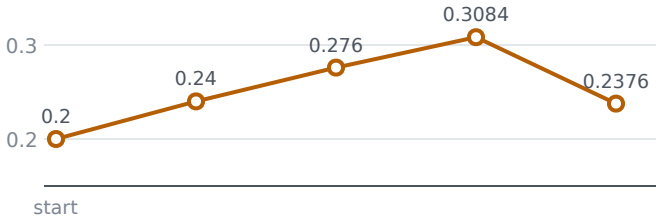
a The last micro-batches of an epoch drag the running mean

batch **PRISMA** **PRISMA** **PRISMA** **AVIRIS-NG** -> validation



~90% of the stored value comes from the last 22 micro-batches (0.9^22 = 0.098) - 0.6% of an epoch

running mean = 0.9 x old + 0.1 x batch mean (channel means: EnMAP 0.2, PRISMA 0.6, AVIRIS-NG -0.4; illustrative)

b Training normalises each sensor by itself; evaluation by one running mean

	EnMAP	PRISMA	AVIRIS-NG
training (batch statistics)	(0.2 - 0.2) = 0.0	(0.6 - 0.6) = 0.0	(-0.4 - (-0.4)) = 0.0
evaluation (running mean 0.2)	(0.2 - 0.2) = 0.0	(0.6 - 0.2) = +0.4	(-0.4 - 0.2) = -0.6

amber: an offset the encoder never saw in training (spread 1 assumed)